Access Path Selection in a Relational Database Management System

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ABSTRACT: In a high level query and data manipulation language such as SQL, requests are stated non-procedurally, without reference to access paths. This paper describes how System R chooses access paths for both simple (single relation) and complex queries (such as joins), given a user specification of desired data as a boolean expression of predicates. System R is an experimental database management system developed to carry out research on the relational model of data. System R was designed and built by members of the IBM San Jose Research Laboratory.

1. Introduction

System R is an experimental database management system based on the relational model of data which has been under development at the IBM San Jose Research Laboratory since 1975 <1>. The software was developed as a research vehicle in relational database, and is not generally available outside the IBM Research Division.

This paper assumes familiarity with relational data model terminology as described in Codd <7> and Date <8>. The user interface in System R is the unified query, data definition, and manipulation language SQL <5>. Statements in SQL can be issued both from an on-line casual-user-oriented terminal interface and from programming languages such as PL/I and COBOL.

In System R a user need not know how the tuples are physically stored and what access paths are available (e.g. which columns have indexes). SQL statements do not require the user to specify anything about the access path to be used for tuple retrieval. Nor does a user specify in what order joins are to be performed. The System R optimizer chooses both join order and an access path for each table in the SQL statement. Of the many possible choices, the optimizer chooses the one which minimizes "total access cost" for performing the entire statement.

This paper will address the issues of access path selection for queries. Retrieval for data manipulation (UPDATE, DELETE) is treated similarly. Section 2 will describe the place of the optimizer in the processing of a SQL statement, and section 3 will describe the storage component access paths that are available on a single physically stored table. In section 4 the optimizer cost formulas are introduced for single table queries, and section 5 discusses the joining of two or more tables, and their corresponding costs. Nested queries (queries in predicates) are covered in section 6.

2. Processing of an SQL statement

A SQL statement is subjected to four phases of processing. Depending on the origin and contents of the statement, these phases may be separated by arbitrary intervals of time. In System R, these arbitrary time intervals are transparent to the system components which process a SQL statement. These mechanisms and a description of the processing of SQL statements from both programs and terminals are further discussed in <2>. Only an overview of those processing steps that are relevant to access path selection will be discussed here.

The four phases of statement processing are parsing, optimization, code generation, and execution. Each SQL statement is sent to the parser. where it is checked for correct syntax. A query block is represented by a SELECT list, a FROM list, and a WHERE tree, containing, respectively the list of items to be retrieved, the table(s) referenced, and the boolean combination of simple predicates specified by the user. A single SQL statement may have many query blocks because a predicate may have one

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